RationGuard - Project Overview

1. What is RationGuard?

RationGuard is an AI-powered system that monitors, tracks, and detects fraud in government ration supply chains. It ensures food grains, oil, sugar, etc. reach the right beneficiaries without leakages, duplication, or corruption. It combines AI-based anomaly detection, a web dashboard, and real-time transaction monitoring for demo simulations.

2. Why this project? (Problem it solves)

Problems in real-world ration distribution:

- Fake or duplicate ration cards / beneficiaries.
- Stock mismanagement – incorrect quantity distribution.
- Leakages or corruption – ration diverted illegally.
- Officials cannot monitor thousands of transactions in real-time.

Impact without RationGuard:

- Wastage of ration.
- Eligible families don't receive their entitlements.
- Corruption continues unchecked.

RationGuard solves this by:

- Automatically flagging suspicious transactions.
- Highlighting fraud-prone centers with heatmaps and alerts.
- Allowing officials to take corrective actions quickly.

3. How does RationGuard work? (Working flow)

- **Data Collection / Simulation:** Uses a simulated dataset of beneficiaries, centers, ration type, and anomalies. In real use, it connects to government databases.
- **AI Anomaly Detection:** Checks for duplicate beneficiaries, wrong quantity, wrong location, and ghost beneficiaries.
- **Backend & API:** Handles data input, AI module calls, stores results, and serves APIs to the frontend.
- **Frontend Dashboard:** Displays transactions, shows alerts, and provides a heatmap of fraud-prone centers.
- **Reports & Actions:** Enables exporting flagged reports for officials to take corrective measures.

4. Why is it important / prize-winning?

- Real-world impact – tackles government ration fraud.
- Technical depth – integrates AI, backend, database, and frontend.
- Demo-ready – judges can see live fraud detection.
- Social relevance – demonstrates how AI improves governance.

Summary:

"RationGuard is an AI system ensuring government ration reaches the right beneficiaries by detecting fraud and leakages in real-time, making governance transparent and efficient."